Scientific Notation

PHP also supports scientific notation (aka standard index form), where a number is expressed as $n \times 10^x$. This notation uses the letter 'e' bounded by numbers on either side. If we wanted to express Ramanujan's number in this form, we'd write 1.729×10^3, which looks like this in PHP:

```
var_dump(1.729E3);
```

```
Output:
float(1729)
```

Note that the value is a float, not an integer.

Converting Bases

In the previous section we saw how to represent numbers in the non-decimal base systems using appropriate prefixes in the literal values and that PHP will always use the decimal notation when casting numbers to strings. If we need to display the non-decimal notations, there are built-in functions for doing that:

```
var_dump(decbin(1729));
var_dump(decoct(1729));
var_dump(dechex(1729));
```

```
Output:
string(11) "11011000001"
string(4) "3301"
string(3) "6c1"
```

The three functions are `decbin()`, `decoct()`, and `dechex()`. Each accepts an integer - or a value that can be cast to an integer - and returns the string representing that integer in the required base.

Correspondingly, `bindec()`, `octdec()`, and `hexdec()` will respectively convert a valid binary, octal, or hex string to an integer:

```
var_dump(bindec('11011000001'));
var_dump(bindec('0b11011000001'));
var_dump(octdec('3301'));
```